

7 **(short)** A serial line carries digital data to a system with input X. The system is required to detect a sequence 010 and give an output Y at the end of the detected sequence. Only non-overlapping sequences should be detected (i.e., the output Y should only be high for the “0” underlined in the input sequence 1010101). J-K bistables are to be used in the design of the system.

- (a) Draw the state diagram and state how many bistables are required. [4]
- (b) Write down the state transition table for the system. [6]